

Unit 6 Pre-Test

Functions & Linear Relationships

Standards assessed: NC.8.F.1–F.5, NC.8.EE.5–EE.6

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.F.1 Identify functions and read slope and y-intercept from an equation.

• I can decide whether a relation is a function and read slope and y-intercept from $y = mx + b$.

Match each vocabulary term to its definition, then answer each question.

– v function	A. Where a graph crosses the y-axis (the value of y when $x = 0$).
– v slope	B. A relation in which each input has exactly one output.
– v y-intercept	C. A point written (x, y) that gives a location on the graph.
– v ordered pair	D. The steepness of a line, found as rise over run.
– v linear function	E. A function whose graph is a straight line.
– v input	F. The x-value that is put into a function.

Slope-intercept form — reference

$$y = mx + b$$

m = slope (rise over run) • b = y-intercept (where $x = 0$)

1. Function?

Is this a function? (1, 2), (2, 4), (3, 6)

Answer: _____

2. Function?

Is this a function? (1, 2), (1, 3), (2, 5)

Answer: _____

3. Slope

In $y = 3x + 5$, what is the slope?

$m =$

4. y-intercept

In $y = 3x + 5$, what is the y-intercept?

$b =$

5. Slope

In $y = -2x + 7$, what is the slope?

$m =$

6. y-intercept

In $y = \frac{1}{2}x - 4$, what is the y-intercept?

$b =$

SCORE **4.0** Target Content

NC.8.F.4 Find slope and y-intercept from two points and from a table.

• *I can find the slope and y-intercept from two points or a table of values.*

Find each slope or y-intercept.

7. Slope

Find the slope through (0, 1) and (2, 5).

$m =$

8. Slope

Find the slope through (1, 3) and (4, 12).

$m =$

9. Slope

Find the slope through (2, 7) and (5, 1).

$m =$

10. y-intercept

A line has slope 4 and passes through (0, -3). Find the y-intercept.

$b =$

11. From a table

12. From a table

x: 0, 1, 2 and y: 5, 8, 11. Find the slope.

m =

x: 0, 1, 2 and y: 5, 8, 11. Find the y-intercept.

b =

SCORE **5.0** Advanced Content

NC.8.F.2 Write equations, compare functions, and interpret slope.

• *I can write a linear equation, compare two functions, and interpret slope as a rate of change.*

Answer each question.

13. Write equation

A line has slope 3 and passes through (0, 2). Write $y = mx + b$.

y =

14. Write equation

A line has slope -4 and passes through (0, -1). Write $y = mx + b$.

y =

15. Compare

Function A: $y = 2x + 3$. Function B: slope 5. Which has the greater slope? (A or B)

Answer:

16. Rate

In $y = 0.5x + 10$, how much does y increase when x goes up by 1?

=

17. Slope

Find the slope through (1, 5) and (3, 11).

m =

18. y-intercept

A line has slope 2 and passes through (3, 11). Find the y-intercept.

b =